

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Easy

Time allowed: 10

Score: /5

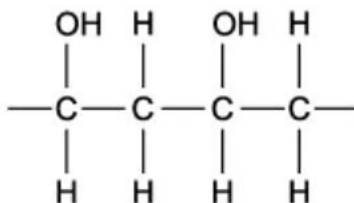
Percentage: /100

Question 1

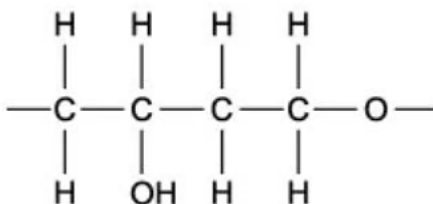
Poly(ethenol) is a polymer used to make dissolvable plastic bags. It can be made by addition polymerisation of $\text{CH}_2=\text{CH}(\text{OH})$.

Which structure shows a polymer chain consisting of two monomer residues?

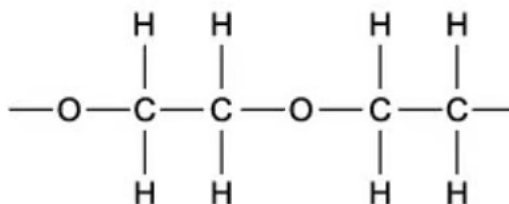
A.



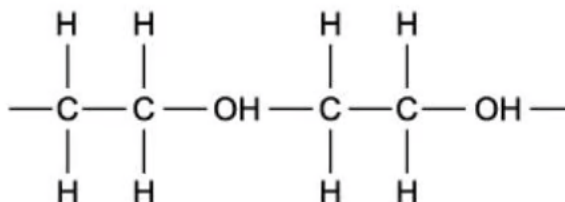
B.



C.



D.



[1 mark]

Question 2

Plastic waste can be incinerated to provide heat for power stations.

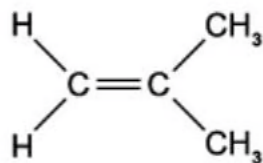
Why is PVC, polyvinylchloride, removed from any plastic waste that is to be incinerated?

- A. It does not burn easily
- B. Its combustion products are harmful
- C. It is easily biodegradable
- D. It destroys the ozone layer

[1 mark]

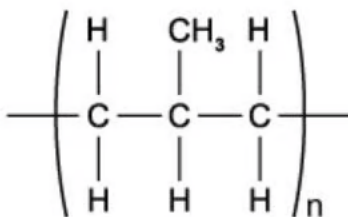
Question 3

The diagram below shows 2-methylpropene, C_4H_8 , a monomer used in the production of synthetic rubber.

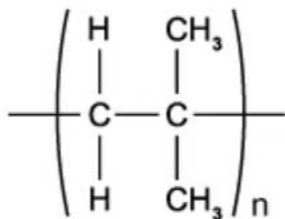


What is the compound formed by 2-methylpropene when it undergoes addition polymerisation?

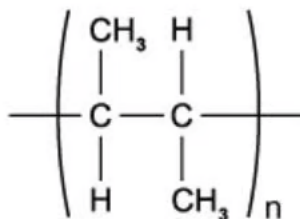
A.



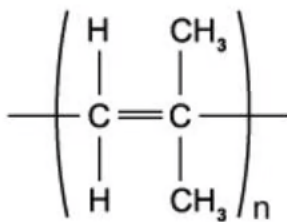
B.



C.



D.



[1 mark]

Question 4

Why is it difficult to dispose of plastics such as poly(alkenes) and PVC?

- 1 PVC produces harmful combustion products
- 2 Poly(alkenes) are non-biodegradable
- 3 Poly(alkenes) are highly flammable

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

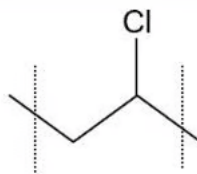
D. 1 only

[1 mark]

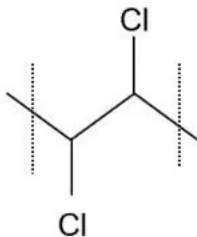
Question 5

What is the repeating unit of PVC?

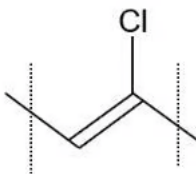
A.



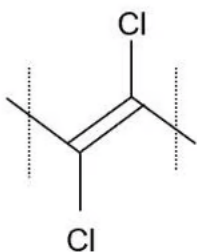
B.



C.



D.



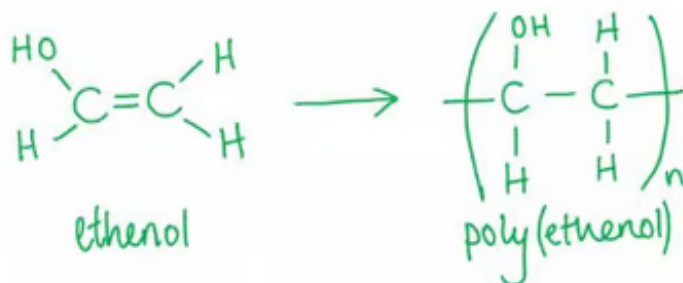
[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- A monomer is one repeating unit of a polymer.
- When ethenol, $\text{CH}(\text{OH})=\text{CH}_2$, is polymerised the carbon carbon **double bond opens** to produce a repeating unit of $\text{CH}(\text{OH})-\text{CH}_2$:



- Therefore, the two of these repeating units looks like structure A.

B is incorrect as the first monomer is correct but the second is not, in **addition polymerisation** there are **only** carbon-carbon bonds connecting monomers together in the main chain.

C is incorrect as in **addition, polymerisation** there are **only** carbon-carbon bonds connecting monomers together in the main chain, the hydroxyl group should remain unmoved and not reacted with.

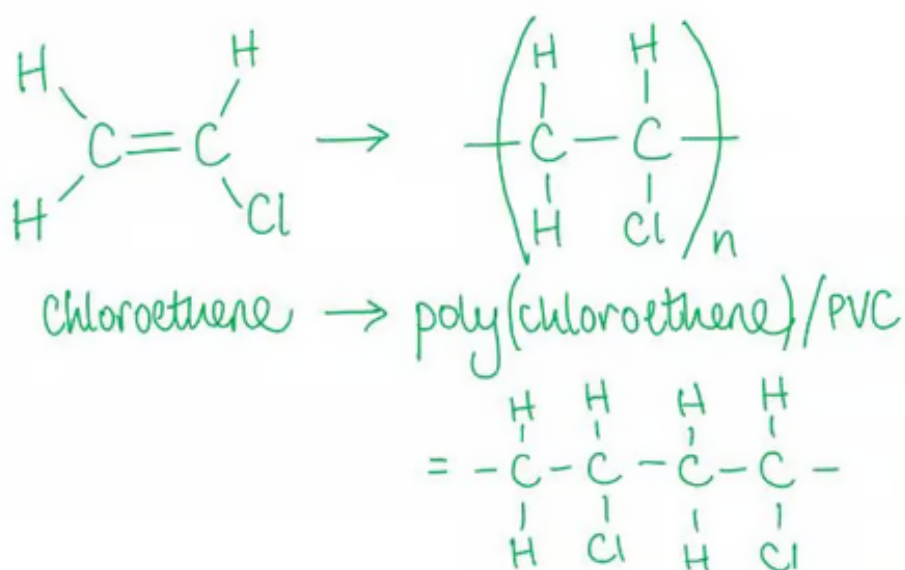
D is incorrect as in **addition, polymerisation** there are **only** carbon-carbon bonds connecting monomers together in the main chain, the hydroxyl group should remain unmoved and not reacted with.

2

The correct answer is **B** because:

- Polyvinylchloride, also known as poly(chloroethene) is a very common polymer

made from the addition polymerisation of chloroethene, $\text{H}_2\text{C}=\text{CHCl}$:



- As polyalkenes can be made from crude oil, the incineration (burning) of these compounds means the energy contained is used for fuel and not 'wasted' by being put in landfill where they cannot decompose.
- When PVC is burned in oxygen **harmful gases** are produced, including **hydrogen chloride**, **dioxins** and **carbon monoxide** all of which are toxic to humans and animals.

***Dioxins** cause problems in reproduction, development, and the immune system.*

***Carbon monoxide** binds to haemoglobin and replaces oxygen which leads to suffocation. **Hydrogen chloride** is an acidic gas which causes irritation to the respiratory tract, eyes and skin.*

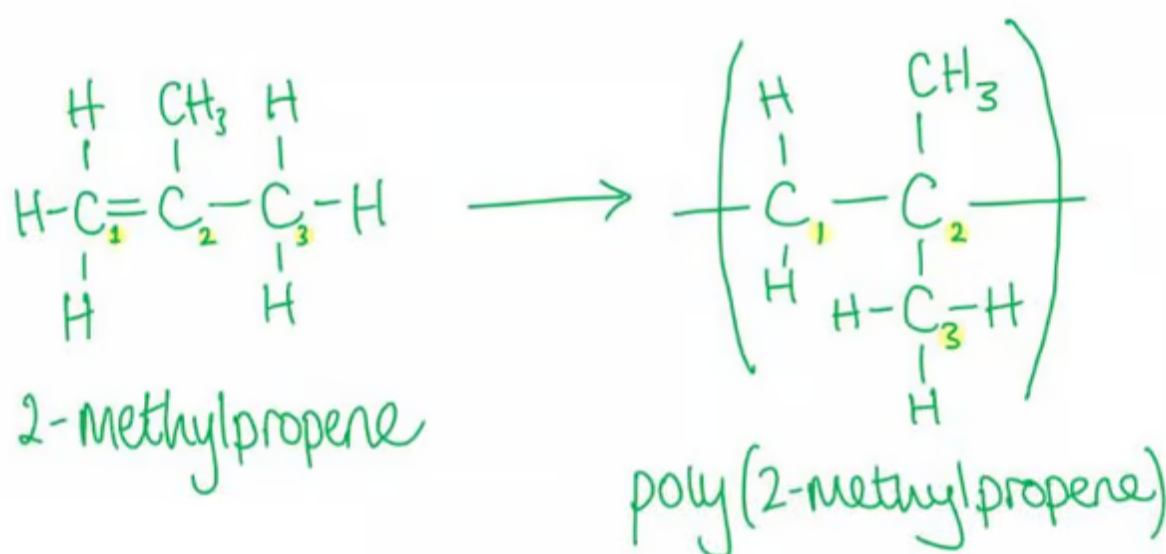
A is incorrect as PVC burns very easily – it is '**highly flammable**'.

C is incorrect as PVC, like other polyalkenes, is **not** biodegradable which means it is not decomposed by bacteria.

D is incorrect as destruction of the ozone layer is caused by **chlorine free radicals** catalysing the formation of oxygen (O_2) from ozone (O_3) – incinerating PVC does **not** produce chlorine free radicals.

The correct answer is **B** because:

- When 2-methylpropene undergoes addition polymerisation its carbon carbon **double bond is broken** to leave a single bond between the carbons and allows **one electron per carbon** to take part in a bond with another monomer.
- The compound's name suggests that the double bond is between carbons 1 and 2 and the additional methyl group is attached to carbon 2.
- Therefore, in the repeating unit the second carbon will have 2 methyl groups attached (one being the 3rd carbon of the propane main chain and the other being the 2-methyl).



A is incorrect as repeating units of addition polymers are **2** carbons long, the other alkyl groups and functional groups are attached to one of the two carbons that shared a double bond.

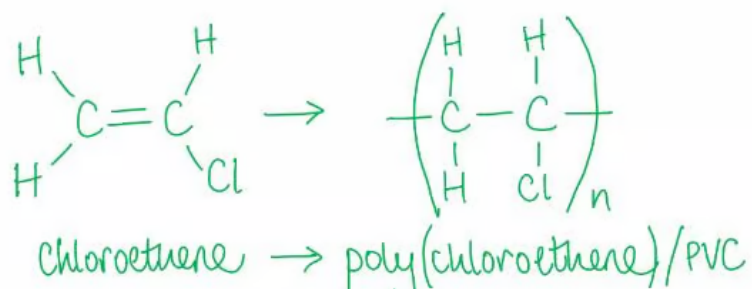
C is incorrect as this is the repeating unit formed in the addition polymerisation of **but-2-ene**.

D is incorrect as during addition polymerisation one of the bonds in the carbon carbon double bond breaks to create a main chain of carbons connected by single bonds.

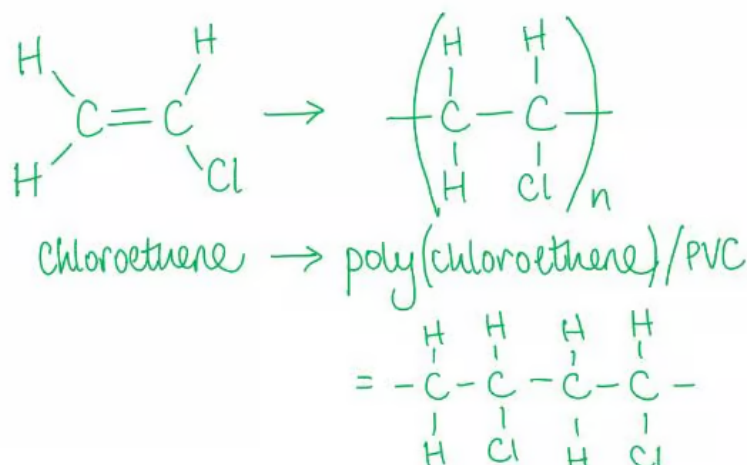
4

The correct answer is **B** because:

- Polyvinylchloride, also known as poly(chloroethene) is a very common polymer made from the addition polymerisation of chloroethene, $\text{H}_2\text{C}=\text{CHCl}$:



...poly(alkenes), also known as poly(chloroethene), is a very common polymer made from the addition polymerisation of chloroethene, $\text{H}_2\text{C}=\text{CHCl}$:



- When PVC is burned in oxygen **harmful gases** are produced, including **hydrogen chloride**, **dioxins** and **carbon monoxide** all of which are toxic to humans and animals.
- Poly(alkenes) are **not** biodegradable which means they are not decomposed by bacteria and hence do not break down for thousands of years when put in landfill.

Dioxins cause problems in reproduction, development, and the immune system.

Carbon monoxide binds to haemoglobin and replaces oxygen which leads to suffocation. Hydrogen chloride is an acidic gas which causes irritation to the respiratory tract, eyes and skin.

A & C are incorrect as poly(alkenes) are highly flammable which means they can be disposed of by incineration.

D is incorrect as statement 2 is also a valid reason – poly(alkenes) do not decompose, which makes them hard to dispose of.

5

The correct answer is **A** because:

- PVC stands for polyvinylchloride, also known as poly(chloroethene).
- PVC is a very common polymer made from the addition polymerisation of chloroethene, $\text{H}_2\text{C}=\text{CHCl}$.
- A repeating unit is the monomer, in this case chloroethene, but the double bond is

broken to form one carbon carbon single bond and an electron for each carbon to form a new bond with the next monomer.

B is incorrect as this is the repeating unit for poly(1,2-dichloroethene).

C & D are incorrect as repeating units do **not** contain **double** bonds.

